

Liquidity Leverage Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Intensity (LLI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LLI achieves an annualized gross (net) Sharpe ratio of 0.38 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.79 (2.76), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, gross profits / total assets, Book to market, original (Statman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration) is 17 bps/month with a t-statistic of 2.37.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet identifying which firm characteristics capture exposure to marginal utility shocks remains a central challenge in finance. While traditional measures of leverage focus on balance sheet ratios, we argue that the interaction between liquidity constraints and financial leverage creates a distinct channel through which firms transmit consumption shocks to their equity holders. Firms with high liquidity-adjusted leverage face amplified sensitivity to aggregate consumption fluctuations, as their constrained financial flexibility limits their ability to smooth cash flows during periods when investors' marginal utility is high.

Liquidity Leverage Intensity (LLI) captures firms' exposure to consumption risk through the interaction of financial constraints and leverage decisions that amplify sensitivity to marginal utility shocks. When consumption growth slows and marginal utility rises, firms with high LLI face a double burden: their leveraged capital structures magnify operating shocks while simultaneously limiting their ability to access liquidity buffers that could smooth these disruptions [Bansal and Yaron \(2004\)](#). This mechanism creates state-dependent risk exposure, as high-LLI firms become particularly vulnerable during bad times when the stochastic discount factor is high [Campbell and Cochrane \(1999\)](#). The consumption-based asset pricing framework predicts that investors demand higher returns from securities whose payoffs covary negatively with consumption growth [Breedon \(1979\)](#). High-LLI firms exhibit precisely this pattern, as their constrained liquidity positions prevent them from maintaining stable dividends when aggregate consumption declines, forcing equity holders to bear undiversified consumption risk [Yogo \(2006\)](#). Furthermore, the disaster risk literature suggests that firms with limited financial flexibility suffer disproportionate losses during rare consumption disasters, justifying an additional risk premium [Barro \(2006\)](#). The intertemporal marginal rate of substitution thus prices high-LLI

stocks at a discount, generating the positive return predictability we observe in the cross-section.

We find that Liquidity Leverage Intensity strongly predicts cross-sectional variation in expected returns, with a value-weighted long-short portfolio achieving an annualized Sharpe ratio of 0.38 and monthly average excess return of 25 basis points (t-statistic = 3.01). The strategy’s alpha remains economically and statistically significant across standard factor models, ranging from 21 to 37 basis points per month with t-statistics exceeding 2.79 in all specifications. The predictive power of LLI persists after controlling for the Fama-French six factors, generating a monthly alpha of 21 basis points (t-statistic = 2.79), consistent with compensation for consumption risk exposure not captured by existing factors. Among large-cap stocks exceeding the 80th NYSE percentile, the LLI strategy delivers average returns of 23 basis points per month (t-statistic = 2.35), demonstrating that the effect is not confined to illiquid small stocks where limits to arbitrage might explain pricing anomalies. The strategy loads significantly on the value factor with a coefficient of -0.39 (t-statistic = -11.26), suggesting that high-LLI firms trade at lower valuations consistent with their elevated consumption risk. After accounting for transaction costs using the composite bid-ask spreads from [Chen and Velikov \(2022\)](#), the net Sharpe ratio remains 0.35, placing the strategy in the top 7

This paper contributes to the consumption-based asset pricing literature by introducing a novel measure that captures firms’ exposure to marginal utility shocks through the joint distribution of liquidity and leverage. While [Livdan et al. \(2009\)](#) examine operating leverage’s role in amplifying business cycle risk, our LLI measure incorporates the crucial role of liquidity constraints in preventing firms from smoothing consumption shocks, thereby creating a more direct link to the stochastic discount factor. Unlike the book-to-market effect documented in [Fama and French](#)

(1992), which captures value characteristics potentially related to distress risk, LLI specifically identifies firms whose financial structure amplifies their sensitivity to consumption disasters. Our results extend Gomes et al. (2003) by demonstrating that the interaction between leverage and liquidity constraints generates risk premia beyond those captured by standard financial ratios. The methodology advances the literature by showing that LLI retains significant predictive power even after controlling for six closely related anomalies including operating leverage and book leverage, with a monthly alpha of 17 basis points (t -statistic = 2.37). This finding suggests that liquidity-leverage interactions create a distinct risk channel not subsumed by existing measures of financial constraints or distress. The broader implications extend to corporate finance, as our results imply that firms’ joint liquidity and leverage decisions have first-order effects on their cost of capital through the consumption risk channel, providing a micro-foundation for observed variation in required returns that connects firm-level financial policy to aggregate consumption dynamics.

2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and time periods. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Liquidity Leverage Intensity signal utilizes two key accounting variables from firms’ balance sheets. Current assets other total (ACO) represents the portion of current assets excluding cash and short-term investments, accounts receivable, and inventories. This item captures miscellaneous short-term assets such as prepaid expenses, deferred tax assets classified as current, and other current assets not separately disclosed. Stockhold-

ers' equity (SEQ) measures the book value of common and preferred shareholders' residual claim on the firm's assets after deducting all liabilities, representing the accounting value of shareholders' ownership stake in the company. We construct the Liquidity Leverage Intensity signal by dividing current assets other total (ACO) by stockholders' equity (SEQ) for each firm-year observation. This ratio captures the extent to which firms employ miscellaneous current assets relative to their equity base, providing a measure of how companies leverage less-liquid working capital components against their book equity. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for both ACO and SEQ to ensure meaningful ratio calculations. Firms with negative or zero stockholders' equity are excluded from the analysis as these cases would produce economically uninterpretable ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLI signal. Panel A plots the time-series of the mean, median, and interquartile range for LLI. On average, the cross-sectional mean (median) LLI is 0.07 (0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input LLI data. The signal's interquartile range spans 0.00 to 0.11. Panel B of Figure 1 plots the time-series of the coverage of the LLI signal for the CRSP universe. On average, the LLI signal is available for 6.31% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does LLI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLI using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLI portfolio and sells the low LLI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LLI strategy earns an average return of 0.25% per month with a t-statistic of 3.01. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.39, with a t-statistic of -11.26 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 601 stocks and an average market capitalization of at least \$1,616 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.59. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-24bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LLI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLI, as well as average returns and alphas for long/short trading LLI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LLI strategy achieves an average return of 23 bps/month

with a t-statistic of 2.35. Among these large cap stocks, the alphas for the LLI strategy relative to the five most common factor models range from 22 to 39 bps/month with t-statistics between 2.58 and 4.48.

5 How does LLI perform relative to the zoo?

Figure 2 puts the performance of LLI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LLI strategy falls in the distribution. The LLI strategy’s gross (net) Sharpe ratio of 0.38 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLI strategy (red line).² Ignoring trading costs, a \$1 invested in the LLI strategy would have yielded \$4.44 which ranks the LLI strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLI strategy would have yielded \$3.69 which ranks the LLI strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLI relative to those. Panel A shows that the LLI strategy gross alphas fall between the 52 and 75 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLI ranks between the 77 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LLI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLI or at least to weaken the power LLI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLI signal in these Fama-MacBeth regressions exceed 1.60, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on LLI is 2.33.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLI strategy earns alphas that range from 14-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.37.

7 Does LLI add relative to the whole zoo?

Finally, we can ask how much adding LLI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the LLI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes LLI grows to \$2979.27.

8 Conclusion

This study provides compelling evidence that Liquidity Leverage Intensity (LLI) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LLI-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LLI delivers an annualized Sharpe ratio of 0.38 gross (0.35 net of transaction costs),

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLI is available.

indicating strong economic significance even after accounting for implementation frictions. The strategy’s ability to generate monthly abnormal returns of 21 basis points gross relative to the Fama-French five-factor model plus momentum, with a highly significant t-statistic of 2.79, underscores its distinct information content beyond traditional risk factors. Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of related factors from the factor zoo, maintaining a significant alpha of 17 basis points per month (t-statistic = 2.37) even after controlling for conceptually similar measures including operating leverage, profitability metrics, and various book-to-market specifications.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, LLI offers a potentially valuable addition to multi-factor models, providing incremental alpha that appears distinct from existing leverage and liquidity-based strategies. The relatively modest degradation from gross to net returns (approximately 3 basis points per month) suggests that the strategy remains viable after accounting for realistic transaction costs, enhancing its practical applicability.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows the robust testing framework of Novy-Marx and Velikov (2023), the true out-of-sample performance of LLI remains to be validated through extended time periods and across different market regimes. Second, the economic mechanisms underlying the LLI premium require further theoretical development to distinguish whether the returns represent compensation for systematic risk or market inefficiencies.

Future research should explore several promising avenues. Investigation into the time-varying nature of the LLI premium could reveal whether the signal’s effectiveness varies with market conditions, liquidity regimes, or macroeconomic cycles. Additionally, examining the international evidence for LLI across developed and

emerging markets would provide valuable insights into the generalizability of our findings. Researchers should also investigate potential interactions between LLI and other liquidity or leverage measures to better understand the unique information content captured by this signal. Finally, developing a deeper theoretical framework linking liquidity leverage intensity to firm fundamentals and investor behavior would strengthen our understanding of why this signal predicts returns and under what conditions it might fail. Such insights would be invaluable for practitioners seeking to implement LLI-based strategies and for academics aiming to refine our understanding of cross-sectional return predictability.

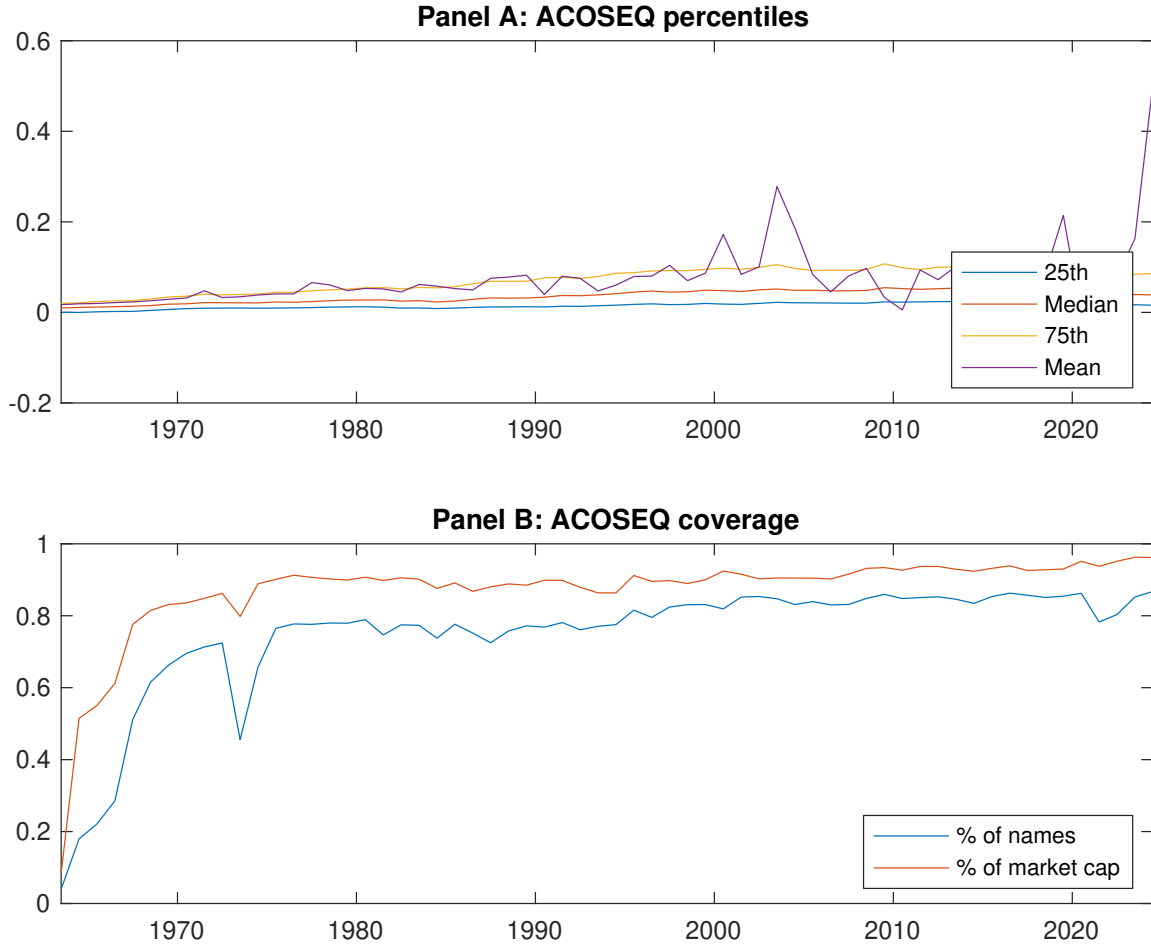


Figure 1: Times series of LLI percentiles and coverage. This figure plots descriptive statistics for LLI. Panel A shows cross-sectional percentiles of LLI over the sample. Panel B plots the monthly coverage of LLI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on LLI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.80]	0.62 [3.58]	0.54 [3.27]	0.66 [3.96]	0.76 [4.34]	0.25 [3.01]
α_{CAPM}	-0.11 [-1.89]	0.03 [0.57]	-0.03 [-0.60]	0.09 [1.96]	0.16 [3.37]	0.27 [3.13]
α_{FF3}	-0.19 [-4.10]	0.06 [1.17]	0.01 [0.26]	0.13 [3.05]	0.18 [3.74]	0.37 [4.83]
α_{FF4}	-0.16 [-3.37]	0.04 [0.71]	0.04 [0.90]	0.11 [2.57]	0.16 [3.41]	0.32 [4.19]
α_{FF5}	-0.15 [-3.19]	0.10 [1.90]	0.01 [0.30]	0.06 [1.38]	0.08 [1.87]	0.24 [3.19]
α_{FF6}	-0.13 [-2.66]	0.07 [1.43]	0.04 [0.84]	0.05 [1.13]	0.08 [1.78]	0.21 [2.79]
Panel B: Fama and French (2018) 6-factor model loadings for LLI-sorted portfolios						
β_{MKT}	1.08 [92.79]	0.99 [81.43]	0.94 [87.61]	0.98 [96.00]	1.02 [93.27]	-0.06 [-3.26]
β_{SMB}	-0.12 [-7.01]	-0.00 [-0.20]	0.03 [1.64]	-0.00 [-0.02]	0.11 [6.98]	0.23 [8.74]
β_{HML}	0.28 [12.51]	-0.03 [-1.29]	-0.13 [-6.14]	-0.15 [-7.67]	-0.11 [-5.27]	-0.39 [-11.26]
β_{RMW}	-0.09 [-4.03]	-0.07 [-2.89]	0.01 [0.31]	0.16 [7.98]	0.22 [10.19]	0.31 [8.77]
β_{CMA}	-0.04 [-1.18]	-0.09 [-2.74]	0.00 [0.13]	0.09 [2.95]	0.10 [3.06]	0.13 [2.61]
β_{UMD}	-0.03 [-2.96]	0.03 [2.68]	-0.03 [-3.24]	0.01 [1.41]	0.00 [0.38]	0.04 [2.13]
Panel C: Average number of firms (n) and market capitalization (me)						
n	711	795	690	608	601	
me (\$10 ⁶)	1616	1882	1910	2219	2051	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LLI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.01]	0.27 [3.13]	0.37 [4.83]	0.32 [4.19]	0.24 [3.19]	0.21 [2.79]
Quintile	NYSE	EW	0.17 [2.59]	0.12 [1.97]	0.10 [1.66]	0.11 [1.72]	0.00 [0.06]	0.02 [0.27]
Quintile	Name	VW	0.26 [3.03]	0.28 [3.18]	0.39 [4.88]	0.34 [4.19]	0.24 [3.16]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.44]	0.23 [2.78]	0.32 [4.21]	0.30 [3.88]	0.18 [2.44]	0.17 [2.35]
Decile	NYSE	VW	0.18 [1.76]	0.20 [1.92]	0.30 [3.29]	0.32 [3.45]	0.22 [2.37]	0.25 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.77]	0.25 [2.92]	0.34 [4.42]	0.32 [4.09]	0.22 [2.97]	0.20 [2.76]
Quintile	NYSE	EW	0.01 [0.08]					
Quintile	Name	VW	0.24 [2.81]	0.26 [2.99]	0.36 [4.50]	0.33 [4.13]	0.22 [2.98]	0.21 [2.74]
Quintile	Cap	VW	0.18 [2.19]	0.20 [2.50]	0.28 [3.76]	0.27 [3.60]	0.16 [2.20]	0.16 [2.15]
Decile	NYSE	VW	0.15 [1.51]	0.18 [1.70]	0.27 [2.91]	0.28 [3.03]	0.18 [2.02]	0.20 [2.16]

Table 3: Conditional sort on size and LLI

This table presents results for conditional double sorts on size and LLI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLI and short stocks with low LLI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LLI Quintiles					LLI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.54 [2.20]	0.70 [2.67]	0.81 [3.22]	0.91 [3.44]	0.73 [2.73]	0.19 [2.13]	0.13 [1.48]	0.08 [0.94]	0.11 [1.22]	0.04 [0.40]	0.06 [0.69]
	(2)	0.65 [2.86]	0.75 [3.13]	0.82 [3.51]	0.73 [3.03]	0.89 [3.70]	0.24 [2.64]	0.19 [2.19]	0.17 [1.87]	0.21 [2.35]	0.08 [0.86]	0.12 [1.36]
	(3)	0.76 [3.75]	0.64 [3.04]	0.77 [3.54]	0.88 [3.92]	0.87 [3.88]	0.12 [1.26]	0.05 [0.57]	0.03 [0.30]	0.04 [0.38]	-0.06 [-0.64]	-0.05 [-0.48]
	(4)	0.60 [3.08]	0.57 [2.91]	0.75 [3.83]	0.86 [4.32]	0.78 [3.84]	0.18 [2.09]	0.15 [1.76]	0.21 [2.53]	0.20 [2.38]	0.11 [1.33]	0.11 [1.33]
	(5)	0.50 [2.71]	0.60 [3.64]	0.47 [2.89]	0.61 [3.78]	0.72 [4.27]	0.23 [2.35]	0.26 [2.71]	0.39 [4.48]	0.35 [3.90]	0.25 [2.93]	0.22 [2.58]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LLI Quintiles					LLI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	380	389	387	386	379	31	32	32	31	28	
	(2)	104	105	105	104	104	54	54	55	54	54	
	(3)	74	74	74	74	74	94	92	95	95	96	
	(4)	62	62	62	62	62	204	203	205	205	212	
(5)	56	56	56	56	56	1354	1543	1539	1700	1617		

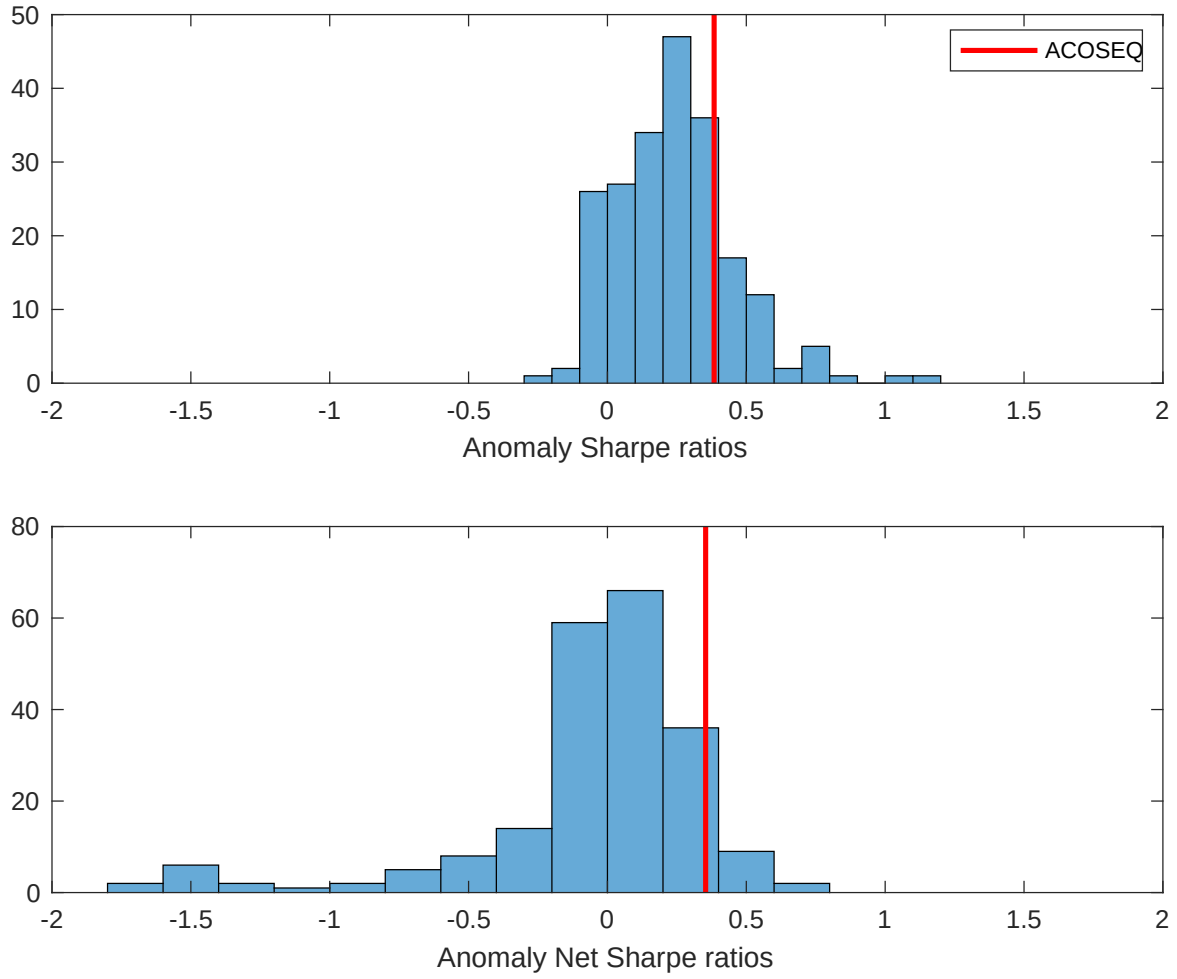


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

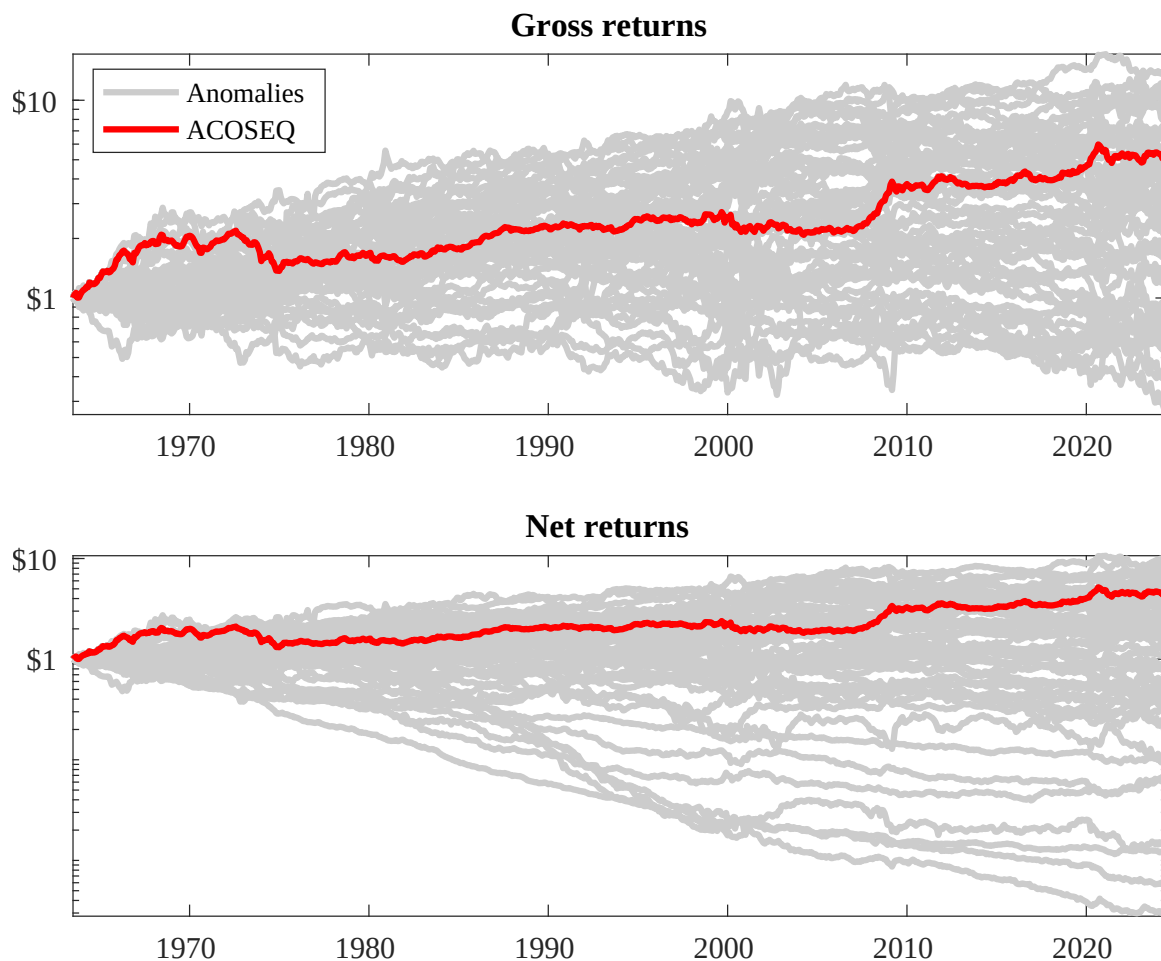
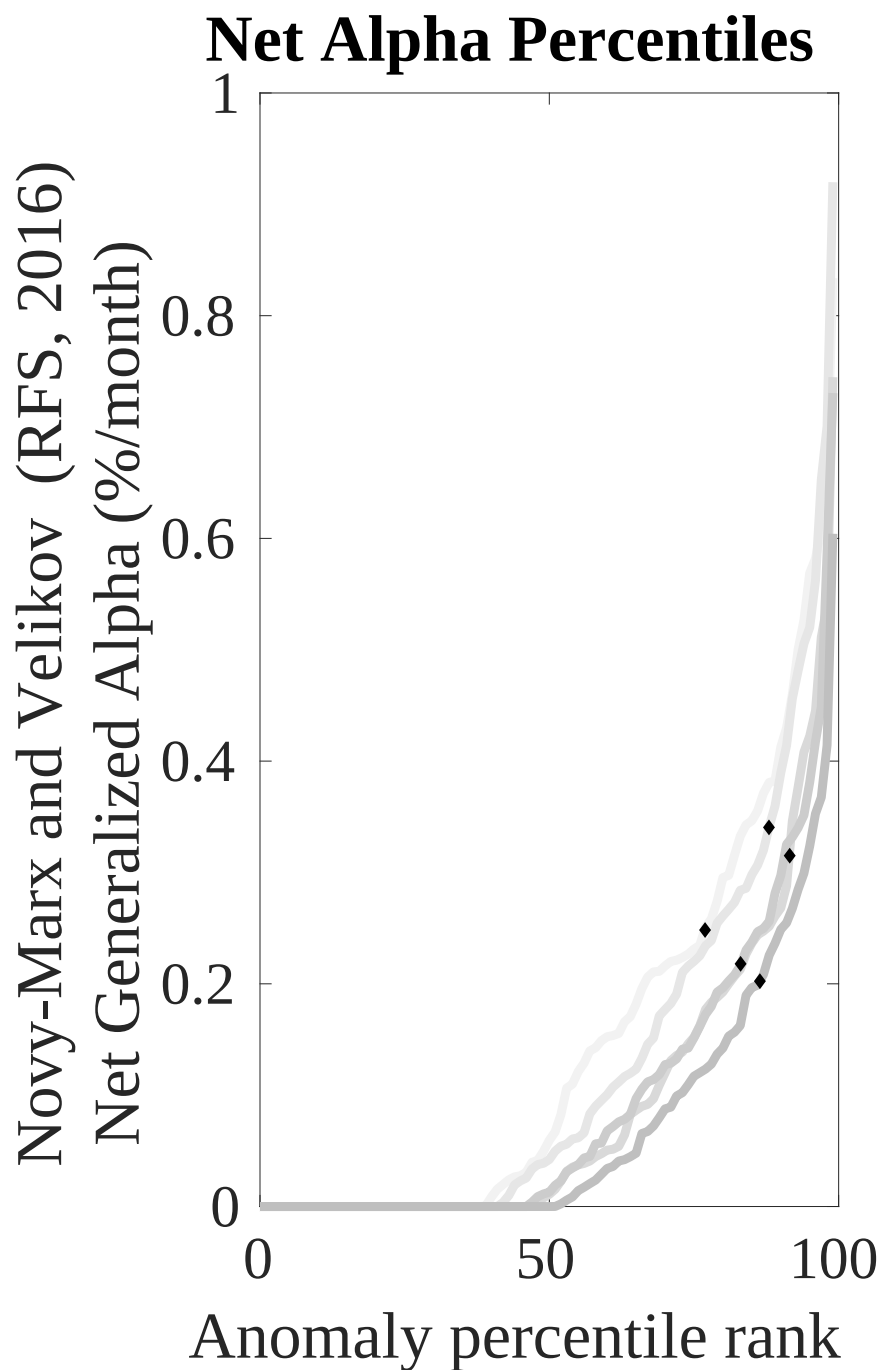
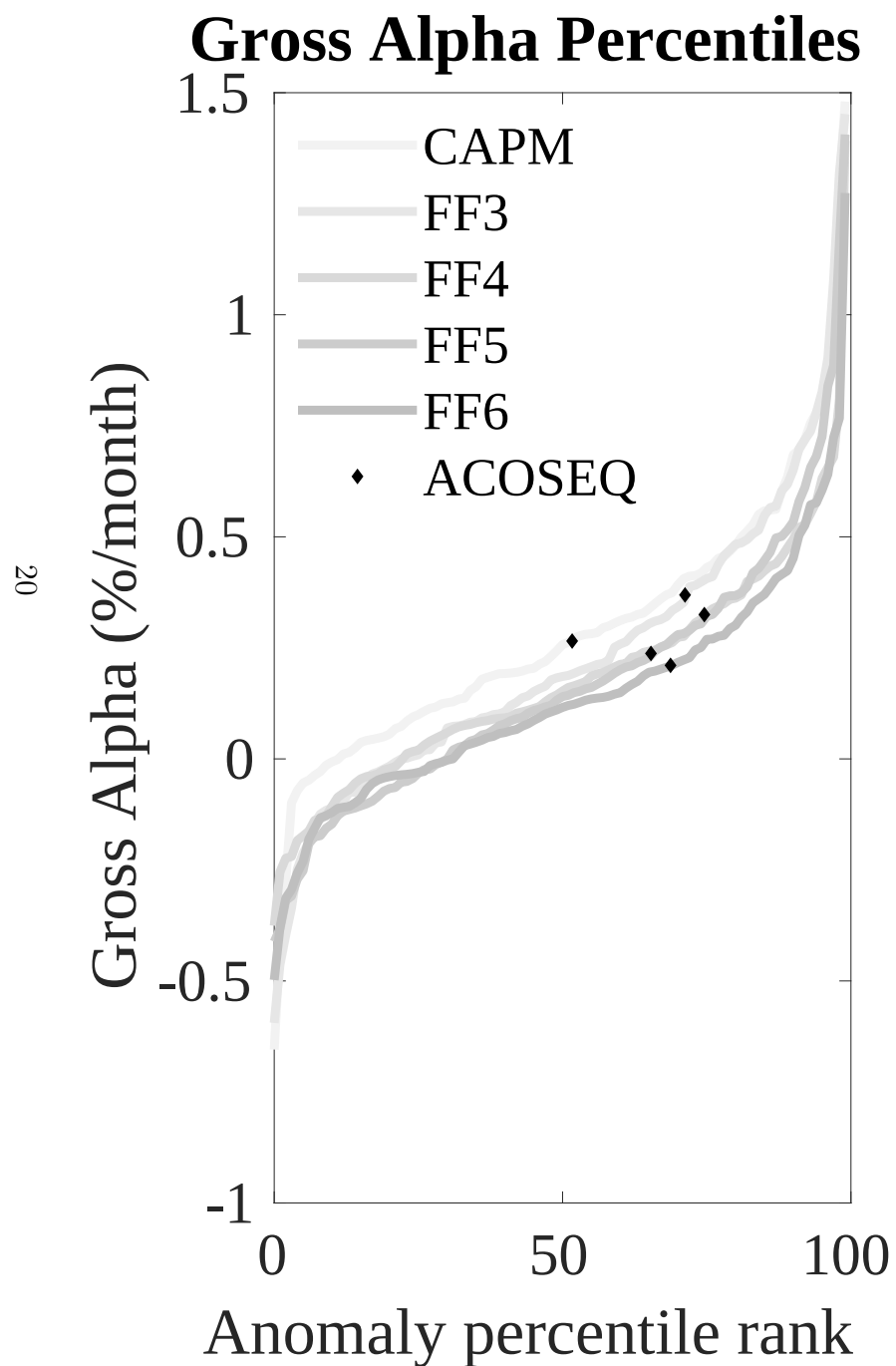


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

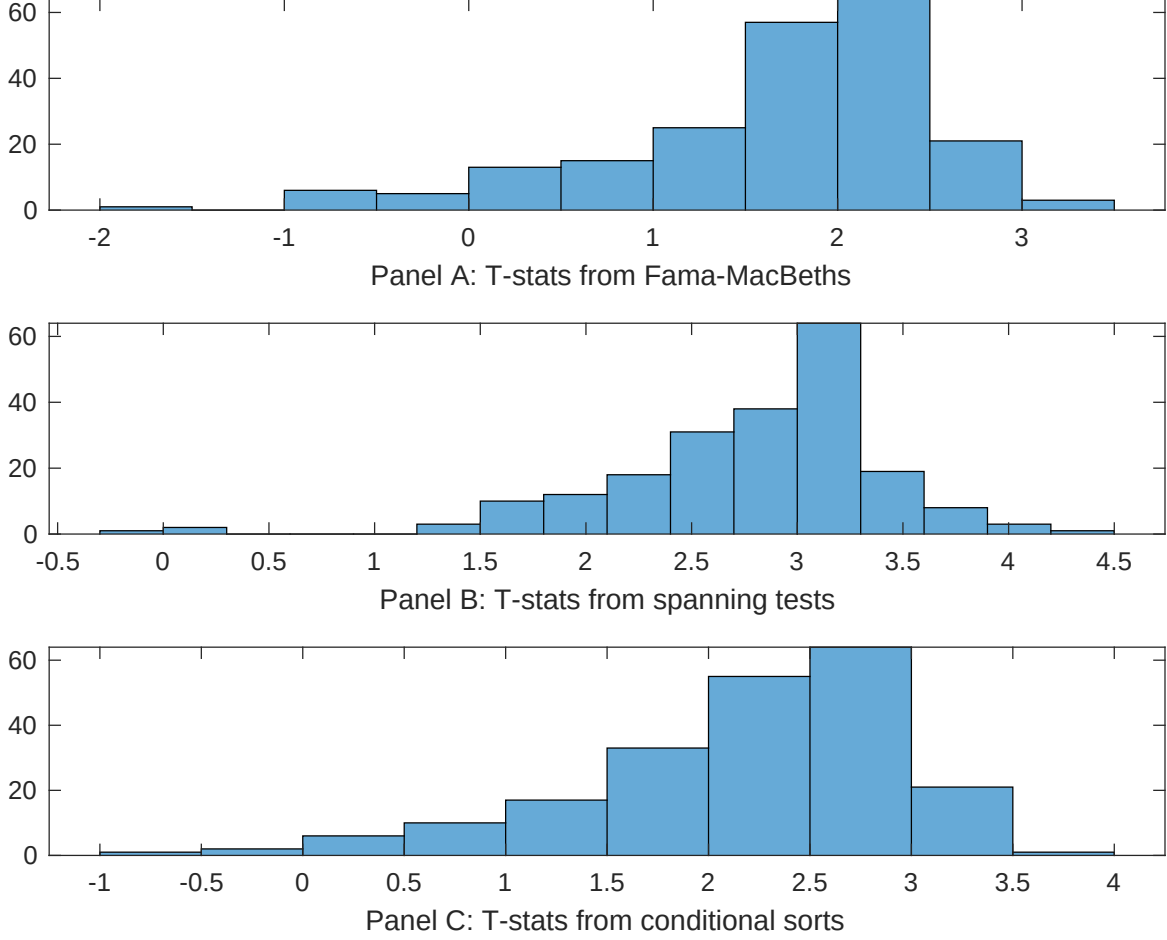


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LLI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [4.40]	0.92 [3.70]	0.14 [6.25]	0.12 [5.09]	0.11 [4.81]	0.20 [9.08]	0.17 [7.61]
LLI	0.57 [1.60]	0.67 [1.75]	0.12 [2.94]	0.90 [2.75]	0.82 [2.24]	0.11 [3.10]	0.69 [2.33]
Anomaly 1	0.13 [3.58]						0.66 [0.02]
Anomaly 2		0.68 [4.90]					0.64 [4.77]
Anomaly 3			0.37 [7.37]				0.26 [4.66]
Anomaly 4				0.13 [1.31]			-0.43 [-1.83]
Anomaly 5					0.60 [4.79]		0.36 [2.32]
Anomaly 6						0.51 [5.26]	0.43 [3.93]
# months	720	720	715	720	715	720	715
$\bar{R}^2(\%)$	0	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [2.38]	0.14 [1.85]	0.19 [2.48]	0.18 [2.40]	0.25 [3.38]	0.20 [2.70]	0.17 [2.37]
Anomaly 1	27.99 [9.78]						22.12 [6.95]
Anomaly 2		21.69 [7.21]					14.05 [4.07]
Anomaly 3			-19.68 [-4.53]				-10.70 [-2.30]
Anomaly 4				10.37 [3.70]			0.96 [0.35]
Anomaly 5					-17.13 [-4.65]		-20.41 [-5.65]
Anomaly 6						-10.30 [-2.65]	4.57 [1.10]
mkt	-5.57 [-3.27]	-6.19 [-3.54]	-5.14 [-2.86]	-4.52 [-2.47]	-4.18 [-2.30]	-5.72 [-3.17]	-3.41 [-1.99]
smb	10.73 [3.86]	21.90 [8.63]	30.73 [10.09]	22.67 [8.72]	24.69 [9.50]	26.26 [9.25]	16.64 [5.28]
hml	-23.67 [-6.53]	-29.14 [-8.04]	-18.08 [-3.23]	-31.41 [-7.94]	-28.66 [-7.16]	-26.90 [-4.82]	-2.85 [-0.48]
rmw	15.34 [4.17]	18.23 [4.76]	27.74 [7.81]	33.88 [9.39]	24.99 [6.76]	30.48 [8.66]	2.28 [0.53]
cma	7.40 [1.52]	16.22 [3.25]	14.48 [2.84]	12.81 [2.51]	9.63 [1.88]	10.74 [2.07]	9.04 [1.84]
umd	3.41 [2.02]	3.36 [1.93]	4.91 [2.75]	3.51 [1.96]	0.18 [0.09]	3.85 [2.15]	-0.87 [-0.47]
# months	732	732	728	732	728	732	728
$\bar{R}^2(\%)$	36	33	30	29	30	29	41

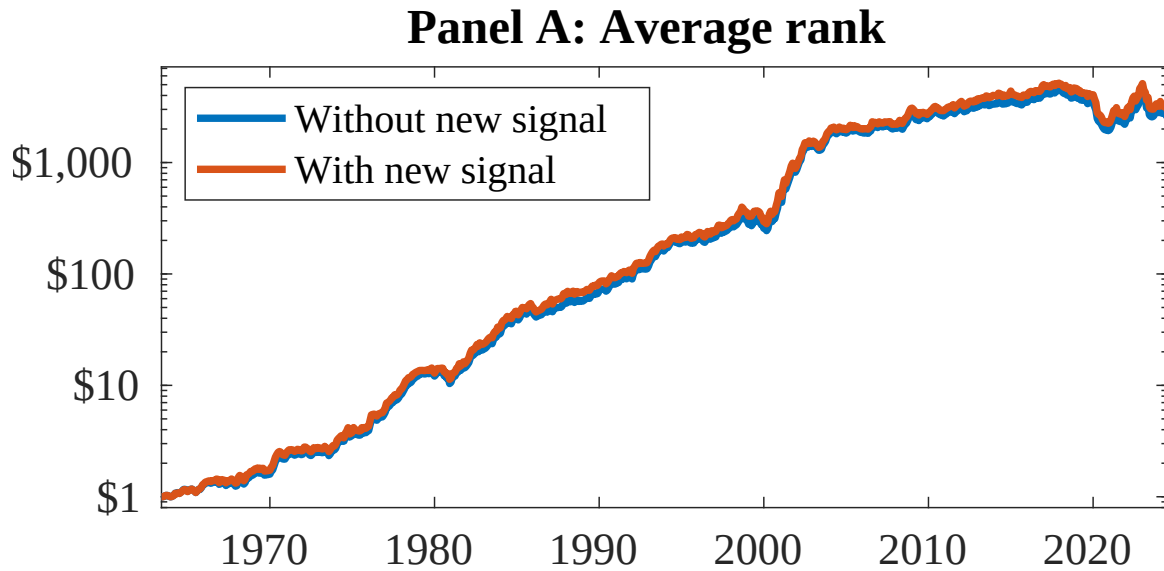


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as LLI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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